

Exercise Sheet 5: Least-Squares Approximation

H By Hand; **C** Computer; **T** Theory; **E** Extra; **A** Advanced.

Recommended: C1; H2.a; C3; H4; 6.a; H7.a; H8; C9; C13; H14.b; H16.a; C17.b.

Polynomial Approximation

- C1.** Use Matlab's `polyfit` function to compute least-squares approximations of degrees 1, 2 and 3 to the following data:

x	0.5	1.0	2.0	2.5	3.0	3.5	4.0
y	1.61	2.13	2.56	2.47	2.26	2.05	1.96

For each case, compute the root-mean-square error of the approximation over the data points, and estimate the value of y when $x = 1.5$.

Answer

```
xs=[0.5,1.0,2.0,2.5,3.0,3.5,4.0]
ys=[1.61,2.13,2.56,2.47,2.26,2.05,1.96]
n=numel(xs)
for deg=1:3,
    cs=polyfit(xs,ys,deg)
    p=@(x)polyval(cs,x);
    erms=sqrt(sum((ys-p(xs)).^2)/n)
    y=p(1.5)
end
```

- H2.** Find the linear least-squares approximation for the following data, compute the root-mean-square error, and estimate the missing value. Graph the data and the approximation.

a.	x_i	0.00	0.25	0.50	0.75	1.00	0.4
	y_i	0.876	1.060	1.525	1.993	2.594	
b.	x_i	1.0	1.1	1.3	1.5	1.9	2.1
	y_i	1.33	1.45	1.70	1.94	2.43	2.67

Answer

- a.** $p_1(x) = 0.0423x + 1.5504$; $E_1 = 0.6218 = 6.2-1$ (2sf), $p_1(0.4) = 1.5673 = 1.567$ (3dp)
b. $p_1(x) = 1.2196x + 0.1109$; $E_1 = 0.0021 = 2.1-3$ (2sf), $p_1(2.0) = 2.5501 = 2.55$ (2dp)

- C3.** Write a Matlab function to compute the coefficients a_0, a_1 of the linear least-squares approximation to data (x_i, y_i) , $i = 1, \dots, n$.

Use your function to compute the linear least-squares approximation to the data in Question 1.

- H4.** Find the least-squares polynomial approximations of degrees 2 and 3 for the data in Question 2, compute the root-mean-square error, and estimate the missing value.

Answer

- a.** $p_2(x) = -0.39223x^2 + 2.12088x + 0.72899$, $E_2 = 0.12804 \approx 1.3-1$, $p_2(x) = 1.51459 \approx 1.515$.
 $p_3(x) = -0.33844x^3 + 1.62971x^2 + 0.44377x + 0.86843$, $E_3 = 0.01199 \approx 1.2-2$, $p_3(0.4) = 1.28503 \approx 1.285$.
b. $p_2(x) = -0.01085x^2 + 1.25329x + 0.08658$, $E_2 = 0.00173 \approx 1.7-3$, $p_2(2.0) = 2.54975 \approx 2.55$.
 $p_3(x) = -0.01005x^3 + 0.03533x^2 + 1.18501x + 0.11902$, $E_3 = 0.00170 \approx 1.7-3$, $p_3(2.0) = 2.54999 \approx 2.55$.

- C5.** Write a Matlab function to compute the coefficients $a_0, a_1, \dots, a_k$ of the polynomial approximation to the linear least-squares approximation to data (x_i, y_i) , $i = 1, \dots, n$.

Use your function to compute the least-squares polynomial approximations of degrees up to 4 for the data in Question 1.

- 6.** Find the linear and quadratic least squares polynomial approximations to f on the indicated interval:

- a.** $f(x) = x^2 - 2x + 3$ on $[0, 1]$; **b.** $f(x) = e^x$ on $[0, 2]$; **c.** $f(x) = \ln x$ on $[1, 3]$.

Answer

- a.** $p_1(x) = -x + 2.8\dot{3}$, $p_2(x) = x^2 - 2x + 3$.
b. $p_1(x) = 3.0000x + 0.1945$, $p_2(x) = 1.4590x^2 + 0.0821x + 1.1672$.
c. $p_1(x) = 0.5281x - 0.4093$, $p_2(x) = -0.1406x^2 + 1.0906x - 0.9239$.

H7. Compute the coefficients of Legendre basis polynomials for the least squares polynomial approximations of degrees 1, 2 and 3 on the interval $[-1, 1]$ for the following functions:

- a.** $f(x) = x^3$ **b.** $f(x) = e^x$ **c.** $f(x) = \ln(x + 2)$

Compute the total square errors. How does the difference in the errors relate to the coefficients?

Answer

- a.** $g_1(x) = g_2(x) = 0.6P_1(x) = 0.6x$,
 $g_3(x) = 0.6P_1(x) + 0.4P_3(x) = 0.6x + 0.4(\frac{5}{2}x^3 - \frac{3}{2}x) = 0.6x + (1.0x^3 - 0.6x) = 1.0x^3$.
 $E_1 = E_2 = 0.0457 = 4.6-2$, $E_3 = 0$.

b.

$$\begin{aligned} c_0 &= (0 + \frac{1}{2}) \int_{-1}^{+1} P_0(x)f(x)dx = \frac{1}{2} \int_{-1}^{+1} e^x dx = \frac{1}{2} [e^x]_{-1}^{+1} \\ &= \frac{1}{2}(e - 1/e) = 1.1752 \\ c_1 &= (1 + \frac{1}{2}) \int_{-1}^{+1} P_1(x)f(x)dx = \frac{3}{2} \int_{-1}^{+1} xe^x dx = \frac{3}{2} [(x-1)e^x]_{-1}^{+1} \\ &= \frac{3}{2}(2/e) = 1.1036 \\ c_2 &= (2 + \frac{1}{2}) \int_{-1}^{+1} P_2(x)f(x)dx = \frac{5}{2} \int_{-1}^{+1} (\frac{3}{2}x^2 - \frac{1}{2})e^x dx = \frac{5}{2} [(\frac{3}{2}x^2 - \frac{6}{2}x + \frac{5}{2})e^x]_{-1}^{+1} \\ &= \frac{5}{2}(e - 7/e) = 0.3578 \\ c_3 &= (3 + \frac{1}{2}) \int_{-1}^{+1} P_3(x)f(x)dx = \frac{7}{2} \int_{-1}^{+1} (\frac{5}{2}x^3 - \frac{3}{2}x)e^x dx = \frac{7}{2} [(\frac{5}{2}x^3 - \frac{15}{2}x^2 + \frac{27}{2}x - \frac{27}{2})e^x]_{-1}^{+1} \\ &= \frac{7}{2}(-5e + 37/e) = 0.0746 \end{aligned}$$

$$g_1(x) = 1.1752P_0(x) + 1.1036P_1(x) = 1.1752 + 1.1036x$$

$$g_2(x) = 1.1752P_0(x) + 1.1036P_1(x) + 0.3578P_2(x) = 1.1752 + 1.1036x + 0.3578(\frac{3}{2}x^2 - \frac{1}{2})$$

$$g_3(x) = 1.1752P_0(x) + 1.1036P_1(x) + 0.3578P_2(x) + 0.0705P_3(x)$$

$$E_1 = \int_{-1}^{+1} (f(x) - g_1(x))^2 dx = 0.052653 = 5.3-2;$$

$$E_2 = \int_{-1}^{+1} (f(x) - g_2(x))^2 dx = 0.001441 = 1.4-3$$

$$= E_1 - \alpha_2 c_2^2 = 0.052653 - \frac{2}{3} \times 0.3578^2 = 0.052653 - 0.051212 = 0.001441$$

$$E_3 = \int_{-1}^{+1} (f(x) - g_3(x))^2 dx = 0.000022 = 2.2-5$$

$$= E_2 - \alpha_3 c_3^2 = 0.001441 - \frac{2}{5} \times 0.0746^2 = 0.001441 - 0.001418 = 0.000022$$

- c.** $g_3(x) = 0.64792P_0(x) + 0.52812P_1(x) - 0.09374P_2(x) + 0.02002P_3(x)$.

In each case, $E_k = E_{k-1} - \alpha_k c_k^2$, where $\alpha_k = \int_{-1}^{+1} P_k(x)^2 dx = 1/(k + \frac{1}{2})$.

H8. The (weighted) least-squares approximations in Legendre and Chebyshev bases for a function f are given by:

a. $p_5(x) = 0.3397 + 1.1505P_1(x) + 0.1711P_2(x) - 1.2405P_3(x) - 0.5983P_4(x) + 0.0816P_5(x)$.

b. $q_5(x) = 0.3069T_0(x) + 0.7059T_1(x) - 0.0405T_2(x) - 0.7514T_3(x) - 0.3054T_4(x) + 0.0421T_5(x)$.

Use the recurrence relations to compute $p_5(x)$ and $q_5(x)$ at $x = 0.7$.

Answer

a. Evaluate Legendre polynomials at x :

$$\begin{aligned} P_0(x) &= 1.00000, \\ P_1(x) &= x = 0.70000, \\ P_2(x) &= \frac{3}{2}xP_1(x) - \frac{1}{2}P_0(x) = \frac{3}{2} \times 0.7 \times 0.70000 - \frac{1}{2} \times 1.00000 = 0.23500, \\ P_3(x) &= \frac{5}{3}xP_2(x) - \frac{2}{3}P_1(x) = \frac{5}{3} \times 0.7 \times 0.23500 - \frac{2}{3} \times 0.70000 = -0.19250, \\ P_4(x) &= \frac{7}{4}xP_3(x) - \frac{3}{4}P_2(x) = \frac{7}{4} \times 0.7 \times -0.19250 - \frac{3}{4} \times 0.2350 = -0.41206, \\ P_5(x) &= \frac{9}{5}xP_4(x) - \frac{4}{5}P_3(x) = \frac{9}{5} \times 0.7 \times -0.41206 - \frac{4}{5} \times -0.19250 = -0.36520. \end{aligned}$$

So

$$\begin{aligned} p_5(x) &= 0.3397 \times 1.00000 + 1.1505 \times 0.70000 + 0.1711 \times 0.23500 \\ &\quad - 1.2405 \times -0.19250 - 0.5983 \times -0.41206 + 0.0816 \times -0.3652 \\ &= 0.33970 + 0.80535 + 0.04021 + 0.23880 + 0.24654 - 0.02980 \\ &= 1.64079 \approx 1.6408 \text{ (4dp)}. \end{aligned}$$

b. $q_5(x) = 1.6258$ (4dp).

C9. Plot the Legendre polynomials P_k for $k = 0, \dots, 8$.

Compute the coefficients c_k of P_k in the best eighth-degree least-squares approximation to the functions given in Question 7, and plot the resulting polynomials p .

Hint: The functions P_k should be computed using the recurrence relation

$$P_k(x) = \left(\left(2 - \frac{1}{k} \right) x P_{k-1}(x) - \left(1 - \frac{1}{k} \right) P_{k-2}(x) \right).$$

For fixed x , you can use this to compute $P_k(x)$ for $k = 0, \dots, 8$ for use in e.g. Romberg integration.

Answer

- a. $g_8(x) = 0.4P_1(x) + 0.6P_3(x)$, $E_8 = 0$.
- b. $g_8(x) = 1.17520119P_0(x) + 1.10363832P_1(x) + 0.35781435P_2(x) + 0.07045563P_3(x) + 0.00996513P_4(x) + 0.000109959P_5(x) + 0.00009945P_6(x) + 0.00000762P_7(x) + 0.00000051P_8(x)$, $E_8 \approx 9-17$.
- c. $g_8(x) = 0.647918P_0(x) + 0.528122P_1(x) - 0.093741P_2(x) + 0.020023P_3(x) - 0.004588P_4(x) + 0.001091P_5(x) - 0.000265P_6(x) + 0.000066P_7(x) - 0.000016P_8(x)$, $E_8 = 1.9-12$.

C10. Plot the Chebyshev polynomials T_k for $k = 0, \dots, 8$.

Compute the coefficients c_k of T_k in the best eighth-degree least-squares approximation using weight $w(x) = 1/\sqrt{1-x^2}$ to the functions given in Question 7, and plot the resulting polynomials p .

Hint: The functions T_k should be computed using the recurrence relation

$$T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x).$$

Plot and compare the linear and quadratic Legendre and Chebyshev approximations. Which give the best *uniform* approximations?

AT11. Construct the *Laguerre polynomials* L_k of degrees up to 4, which are orthogonal on the interval $[0, \infty)$ with respect to the weight function $w(x) = e^{-x}$.

Answer

$$\begin{aligned} L_0(x) &= 1, \\ L_1(x) &= (x-1)L_0(x) = x-1, \\ L_2(x) &= (x-3)L_1(x) - L_0(x) = x^2 - 4x + 2, \\ L_3(x) &= (x-5)L_2(x) - 4L_1(x) = x^3 - 9x^2 + 18x - 6. \end{aligned}$$

E12. Use the Laguerre polynomials calculated to compute the least squares polynomials of degree $1, \dots, 4$ on the interval $[0, \infty)$ with respect to the weight function $w(x) = e^{-x}$ for the following functions.

- a. $f(x) = x^2$ b. $f(x) = e^{-x}$ c. $f(x) = x^3$ d. $f(x) = e^{-2x}$

Fourier Series

C13. Use Matlab to compute the coefficients of the Fourier approximation $s_5(x)$ of degree $n = 5$ to the function $f(x) = (\cos x)/(2 + \sin x)$. Plot the graph of the function f and the Fourier approximations s_1, s_2, s_3 .

The *energy* of the signal in the k -th mode is $a_k^2 + b_k^2$. Comment on the how energy in the k^{th} mode for $k = 1, \dots, n$ relates to the form of the function.

Hint: You may wish to write a function `fourieval(as,bs,x)`, which evaluates the Fourier series with coefficients a_k, b_k at the point x . Alternatively, you could use arguments `(a0,as,bs,x)`.

Answer $s_5(x) = 0.5359 \cos(x) - 0.1436 \sin(2x) - 0.385 \cos(3x) + 0.0103 \sin(4x) + 0.0028 \cos(5x)$

H14. Find the Fourier approximations $s_n(x)$ on $[-\pi, \pi]$ for $n = 1, 2, 3$ and the following functions.

- a. $f(x) = x \sin x$; b. $f(x) = e^x$.

Compute the total-square-errors and the mean-square-errors. How does the difference in the total-square-errors relate to the coefficients?

Hint: You can evaluate the integrals by parts.

Answer

- a. $s_1(x) = 1 - \frac{1}{2} \cos(x)$, $s_2(x) = 1 - \frac{1}{2} \cos(x) - \frac{2}{3} \cos(2x)$,
 $s_3(x) = 1 - \frac{1}{2} \cos(x) - \frac{2}{3} \cos(2x) + \frac{1}{4} \cos(3x)$.
 $E_1 = 1.69695$, $E_2 = 0.29978$, $E_3 = 0.10343$

b.

$$\int e^x \cos(kx) dx = \frac{e^x (\cos(kx) + k \sin(kx))}{1 + k^2}, \quad \int e^x \sin(kx) dx = \frac{e^x (-k \cos(kx) + \sin(kx))}{1 + k^2},$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{+\pi} e^x \cos(kx) dx = \frac{(-1)^k}{1 + k^2} \frac{e^\pi - e^{-\pi}}{\pi}, \quad b_k = \frac{(-1)^{k-1} k}{1 + k^2} \frac{e^\pi - e^{-\pi}}{\pi}.$$

$$s_3(x) = 3.6761 - 3.6761 \cos(x) + 3.6761 \sin(x) \\ + 1.4704 \cos(2x) - 2.9409 \sin(2x) - 0.7352 \cos(3x) + 2.2056 \sin(3x).$$

$$E_1 = \int_{-\pi}^{+\pi} (f(x) - s_1(x))^2 dx = 97.9286 \approx 98.$$

$$E_2 = \int_{-\pi}^{+\pi} (f(x) - s_2(x))^2 dx = 63.9654 \approx 62.$$

$$= E_1 - \pi(a_2^2 + b_2^2) = 97.9286 - \pi(1.4704^2 + 2.9409^2) = 97.9286 - 33.9633$$

$$E_3 = \int_{-\pi}^{+\pi} (f(x) - s_3(x))^2 dx = 46.9838 \approx 46$$

$$= E_2 - \pi(a_3^2 + b_3^2) = 63.9654 - \pi(0.7352^2 + 2.2056^2) = 63.9654 - 16.9816$$

In all cases, $E_k = E_{k-1} - \pi(a_k^2 + b_k^2)$.

E15. Determine the Fourier approximations $s_n(x)$ on the interval $[-\pi, \pi]$ for the functions given in Question 14 by computing $2m$ data points $y_j = f(x_j)$ with $x_j = j\pi/m$, using $m = 4, 8$ and $n = 3$. Compare your answers with those found previously.

H16. Find the Fourier approximations $S_2(x)$ on $[-\pi, \pi]$ for the following data:

a.	x_i	$-\pi$	$-2\pi/3$	$-\pi/3$	0	$\pi/3$	$2\pi/3$	π	$\pi/2$		
	y_i	0.54	0.37	0.69	1.87	2.69	1.44	0.54			
b.	x_i	$-\pi$	$-3\pi/4$	$-\pi/2$	$-\pi/4$	0	$\pi/4$	$\pi/2$	$3\pi/4$	π	$\pi/3$
	y_i	0.49	0.81	0.98	0.96	0.75	0.42	0.08	0.14	0.49	

Use your approximations to estimate the missing value.

Answer

a.

$$a_0 = \frac{1}{3} \sum_{j=-3}^2 y_j = \frac{1}{3} (0.54 + 0.37 + 0.69 + 1.87 + 2.69 + 1.44) \\ = 2.5333$$

$$a_1 = \frac{1}{3} \sum_{j=-3}^2 y_j \cos(x_j) \\ = \frac{1}{3} (0.54 \cos(-\pi) + 0.37 \cos(-2\pi/3) + 0.69 \cos(\pi/3) + 1.87 \cos(0) + 2.69 \cos(\pi/3) + 1.44 \cos(2\pi/3)) \\ = \frac{1}{3} (0.54 \times (-1.0) + 0.37 \times (-0.5) + 0.69 \times 0.5 + 1.87 \times 1.0 + 2.69 \times 0.5 + 1.44 \times (-0.5)) \\ = 0.7050$$

$$a_2 = \frac{1}{3} \sum_{j=-3}^2 y_j \cos(2x_j) \\ = \frac{1}{3} (0.54 \cos(-2\pi) + 0.37 \cos(-4\pi/3) + 0.69 \cos(2\pi/3) + 1.87 \cos(0) + 2.69 \cos(2\pi/3) + 1.44 \cos(4\pi/3)) \\ = \frac{1}{3} (0.54 \times 1.0 + 0.37 \times (-0.5) + 0.69 \times (-0.5) + 1.87 \times 1.0 + 2.69 \times (-0.5) + 1.44 \times (-0.5)) \\ = -0.0617$$

$$b_1 = \frac{1}{3} \sum_{j=-3}^2 y_j \sin(x_j) \\ = \frac{1}{3} (0.54 \times 0.0 + 0.37 \times (-0.8660) + 0.69 \times (-0.8660) + 1.87 \times 0.0 + 2.69 \times 0.8660 + 1.44 \times 0.8660) \\ = 0.8862$$

$$b_2 = \frac{1}{3} \sum_{j=-3}^2 y_j \sin(2x_j) \\ = \frac{1}{3} (0.54 \times 0.0 + 0.37 \times 0.8660 + 0.69 \times (-0.8660) + 1.87 \times 0.0 + 2.69 \times 0.8660 + 1.44 \times (-0.8660)) \\ = 0.2685$$

$$s_2(x) = 2.5333/2 + 0.7050 \cos(x) + 0.8862 \sin(x) - 0.0617 \cos(2x) + 0.2685 \sin(2x)$$

$$\mathbf{b.} \quad s_2(x) = 1.1575/2 + 0.1410 \cos(x) + 0.0450 \sin(x) - 0.4389 \cos(2x) + 0.0325 \sin(2x)$$

C17. Use Matlab to compute the coefficients of the Fourier approximations to the data (x_i, y_i) below:

a.	x_i	$-\pi$	$-3\pi/4$	$-\pi/2$	$-\pi/4$	0	$\pi/4$	$\pi/2$	$3\pi/4$
	y_i	0.17647	0.33194	0.47059	0.88653	0.76471	0.60924	0.47059	0.05464
b.	x_i	$-\pi$	$-3\pi/4$	$-\pi/2$	$-\pi/4$	0	$\pi/4$	$\pi/2$	$3\pi/4$
	y_i	0.00000	0.06678	0.21630	0.57983	1.00000	0.57983	0.21630	0.06678

Comment on your results.

Answer

$$\mathbf{a.} \quad a_0 = 0.94118, a_1 = 0.34314, a_2 = 0, a_3 = -0.04902, a_4 = 2.5-6 \approx 0, \\ b_1 = -0.09804, b_2 = 2.3-6 \approx 0, b_3 = 0.09804.$$

The $\cos/\sin(2x)$ and $\cos(4x)$ terms are (approximately) zero, so these modes are not present.

$$\mathbf{b.} \quad a_0 = 0.68145, a_1 = 0.43139, a_2 = 0.14185, a_3 = 0.06861, a_4 = 0.034845, b_1 = b_2 = b_3 = 0.$$

The data comes from an even function, so the $\sin(kx)$ terms are zero.

E18. Compute the coefficients of the Fourier series of the *sawtooth wave*, which is the 2π -periodic function satisfying

$$f(x) = x \text{ for } x \in [-\pi, +\pi).$$

Write down the general form of the least-squares approximation s_n .

Plot s_n for $n = 1, 2, 3, \dots$. What happens to the least-squares error as $n \rightarrow \infty$? What happens to the uniform error?

By computing $\int_0^\pi f(x)^2 dx$ and $\int_0^\pi s(x)^2 dx$, find a formula for $\sum_{k=1}^\infty 1/k^2$.

Answer Use the formula

$$\int_0^\pi f(x)^2 dx = \int_0^\pi s(x)^2 dx = \frac{\pi}{4} a_0^2 + \frac{\pi}{2} \sum_{k=1}^\infty (a_k^2 + b_k^2)$$

Since f is odd, $a_k = 0$. Compute

$$\begin{aligned} b_k &= \frac{1}{\pi} \int_{-\pi}^{+\pi} x \sin(kx) dx = \frac{2}{\pi} \int_0^{\pi} x \sin(kx) dx \\ &= \frac{2}{\pi} \left[-x \cos(kx)/k + \sin(kx)/k^2 \right]_0^{\pi} = \frac{2}{\pi} (-1)^{k-1} \pi / 2k = (-1)^{k-1} 2/k. \end{aligned}$$

Square integrals

$$\begin{aligned} \int_0^{\pi} f(x)^2 dx &= \int_0^{\pi} x^2 dx = [x^3/3]_0^{\pi} = \pi^3/3. \\ \int_0^{\pi} s(x)^2 dx &= \frac{\pi}{2} \sum_{k=1}^{\infty} b_k^2 = \frac{\pi}{2} \sum_{k=1}^{\infty} (2/k)^2 = 2\pi \sum_{k=1}^{\infty} 1/k^2. \end{aligned}$$

Hence

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

AC19. Write a Matlab function to compute the discrete Fourier transform using the fast Fourier transform method. Compute the trigonometric interpolating polynomial of degree 4 and 6 on $[-\pi, \pi]$ using $2m = 2^6$ and 2^8 data points for the following functions.

a. $f(x) = \pi(x - \pi)$

b. $f(x) = |x|$

c. $f(x) = x \cos x^2$

E20. Compute the Fourier coefficients b_m up to $m = 15$ for the wavelet

$$f(x) = \exp(3 \cos(x)) \sin(5x).$$

Comment on how the coefficients vary. Which is the largest coefficient?